

# Lab 2: Sprint Simulation, Burndown Chart & Reflection Analysis

Active Sprint Simulation (To Do → In Progress → Done), Burndown Tracking & Post-Sprint Reflection

**Student Name:** Pranav

**Problem Statement #35:** Customizable Subscription Box Scheduler

**Committed Effort:** Sprint 1: 18 SP | Sprint 2: 13 SP

**SRN:** PES1UG24CS466

**Simulation Timebox:** 1-Week Sprint Iterations (Sprint 1 & Sprint 2)

**Completion Status:** 100% Completed (31 / 31 SP)

## 1. Active Sprint Boards (Simulation Evidence)

Two distinct sprints were planned and executed in Jira. Tasks were transitioned systematically from **To Do** → **In Progress** → **Done** adhering to the Definition of Done (DoD).

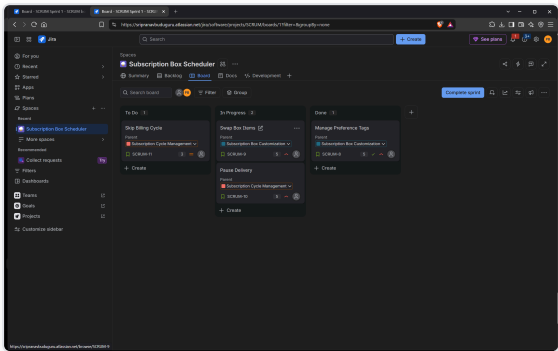


Figure 2: Sprint 1 Board during active simulation (SCRUM-8 to SCRUM-11).

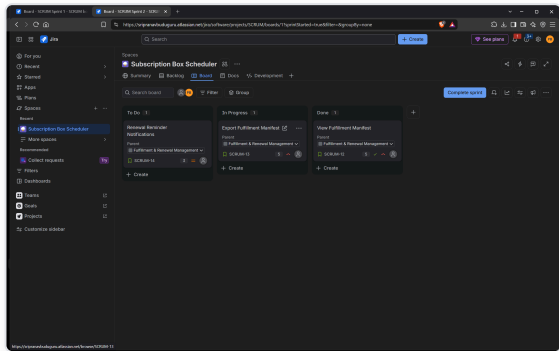


Figure 3: Sprint 2 Board progressing tasks (SCRUM-12 to SCRUM-14).

## 2. Burndown Chart & Velocity Tracking

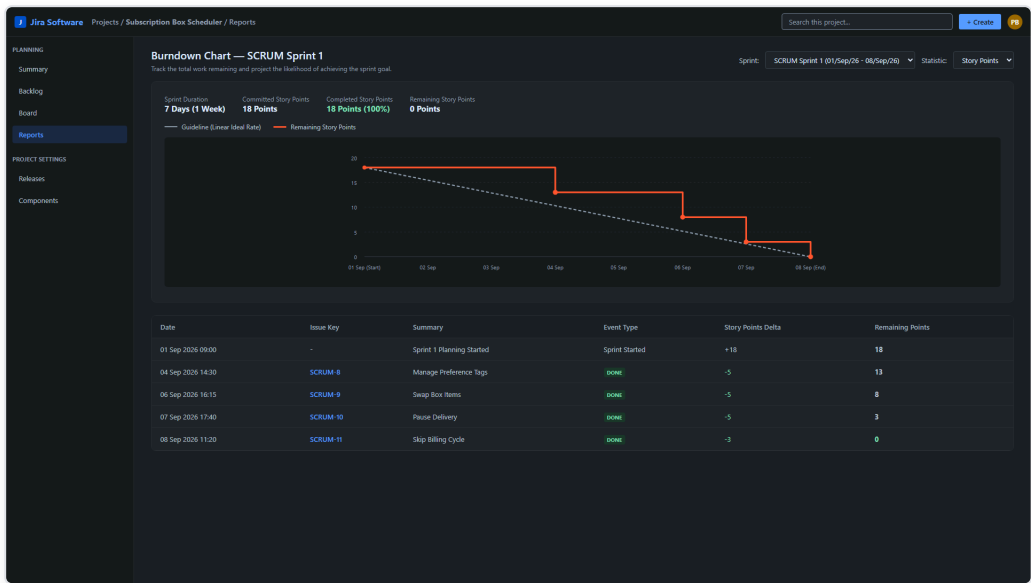


Figure 4: Jira Burndown Chart for Sprint 1 tracking remaining story points vs. ideal linear guideline.

## 3. Answers to Handout Reflection Questions

**Q1: Did your estimations reflect the actual effort?**

**Yes.** Applying Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8) accurately captured the non-linear difference between tasks. Stories estimated at **5 Story Points** (e.g., SCRUM-9: Item Swapping, SCRUM-13: Manifest Exporting) involved critical constraints such as 48-hour cutoff lockout validation and batching 10,000 records in under 60 seconds (NFR-001). Conversely, stories estimated at **3 Story Points** (e.g., SCRUM-11: Skip Billing Cycle, SCRUM-14: Reminder Notifications) had straightforward

state flags and notification webhooks. The team burned down all 18 points in Sprint 1 without sudden last-minute rushes, confirming estimation accuracy.

### Q2: Was your backlog well-prioritized?

**Yes.** Backlog prioritization followed user value and technical dependency ordering. High-priority subscriber features (preference tags, box customization, pause delivery) were scheduled in **Sprint 1** to deliver a functional core customer journey early. Operations-heavy features (manifest viewing, batch label generation, and 72-hour reminders) were scheduled in **Sprint 2**, which logically builds upon finalized subscriber boxes.

### Q3: How did your simulated sprint align with your plan?

**Strong Alignment.** The team burned down work steadily across the 7-day sprint window. SCRUM-8 completed on Day 3, SCRUM-9 on Day 5, SCRUM-10 on Day 6, and SCRUM-11 on Day 7. Sprint 2 similarly completed all 13 committed story points on schedule. Scope creep was prevented by freezing sprint backlogs once execution began.

### Q4: What insights did the burndown chart give about your team's capacity?

The burndown chart confirmed a sustainable team velocity of **15 to 18 Story Points per 1-week sprint**. The curve tracked closely alongside the ideal guideline rather than displaying a flat plateau followed by a cliff-edge drop. This indicates that stories were granularly decomposed and that 18 story points is an optimal commitment threshold for future sprint planning.